THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6

Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 126	Find the breadth first.
2	**(d)** $3\frac{1}{6}$	Convert to improper fractions or borrow.
3	**(c)** 2	A point is inside if its distance from O is less than the radius.
4	**(a)** 18	Lines of symmetry = $360 \div$ angle.
5	**(a)** -13	Add step by step.
6	**(a)** 144	Area becomes 4 times.
7	**(c)** 1	Use the LCM of all denominators.
8	**(c)** 180	Triangle angle sum.
9	**(b)** 300	Multiply.
10	**(a)** 8	Add the opposites.
11	**(c)** 4800	Area $\times$ rate.
12	**(c)** 60	Divide by 3, multiply by 5.
13	**(b)** 130	Linear pair.
14	**(b)** A square	Only the square among these has 4.
15	**(a)** 0	Opposites cancel.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$l + b = 35$, so breadth $= 11$ cm (1); Area $= 264$ cm^2 (1).	1+1
17	Convert to improper fractions (1); $3\frac{1}{4}$ (1).	1+1
18	Width $= 9 - 4 = 5$ cm (1); Inner diameter $= 8$ cm; outer diameter $= 18$ cm (1).	1+1
19	Sides $= 360 \div 40 = 9$ (1); Order $= 9$ (1).	1+1
20	Add the first two: 7 (1); Then add (-9): -2 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Inner lawn: 26 m by 16 m (1); Inner area 416 m^2 (1); Path area $= 600 - 416 = 184$ m^2 (1).	1+1+1
22	Ploughed $=$\frac{1}{4}+\frac{1}{3}$= \frac{7}{12}$ (1); Remaining $= 1 - \frac{7}{12}$ (1); $= \frac{5}{12}$ (1).	1+1+1
23	Area of square $= 36$ cm^2 (1); One triangle is one quarter: 9 cm^2 (1); Angles at centre: four right angles, 360° (1).	1+1+1
24	Angle $= 360 \div 5 = 72^\circ$ (1); After 3 turns: 216° (1); Turns for full rotation $= 5$ (1).	1+1+1
	OR: Start at 0; after 3 down: -3; After 5 up: 2 (1); After 4 down: -2 (1); Total floors moved $= 12$ (1).	1+1+1
25	First area $= 80$ cm^2 (1); Second area $= 144$ cm^2 (1); Larger: second, difference $= 64$ cm^2 (1).	1+1+1
	OR: Mixed to improper: $\frac{5}{2}$ (1); Add $\frac{1}{3}$ (1); Subtract $\frac{1}{4}$: result $2\frac{7}{12}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Hall in cm: 1000 by 800 (1); Tiles along each side: 20 and 16 (1); Number of tiles $= 320$ (1); Cost $=$ ₹12800 (2).	1+1+1+2
	OR: Books: \frac{1}{4}$\times 120 = 30$ (1); Rest $= 90$ (1); Bag: 30 (1); Left $= 60$ (2).	1+1+1+2
27	Draw AB = 4 cm (1); Make right angles at A and B (compass method) (1); Mark AD and BC = 4 cm and join DC (1); Half diagonal OA $= 3$ cm (1); Diagonals meet at 90° (1).	1+1+1+1+1
	OR: (a) 5 (1); (b) 72° (1); (c) 5 (1); (d) 144° is 2 turns so it leaves the pattern unchanged (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 9	Perimeter of square $=4\times$ side.
2	**(c)** $\frac{5}{8}$	Fraction left $=1-$ fraction spent.
3	**(d)** 5	Outer radius minus inner radius.
4	**(a)** 60	The square fits after every multiple of 90°.
5	**(c)** 17	Highest minus lowest.
6	**(b)** 100	Side of square $=$ perimeter $\div 4$.
7	**(d)** $\frac{3}{8}$	Take the average of the two fractions.
8	**(d)** 8	Side $=$ perimeter $\div 4$.
9	**(a)** 10	Add.
10	**(a)** -30	Subtract the descent.
11	**(a)** 256	Inner side $= a - 2 \times 2$.
12	**(c)** 17	Whole times denominator plus numerator.
13	**(a)** 24	4 diameters.
14	**(a)** 5	Count the letters with a fold line.
15	**(d)** -7 and 12	Check each pair.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Area $= 96$ m^2 (1); Cost $= 96 \times 75 =$ ₹7200 (1).	1+1
17	Number $+$\frac{1}{2} of number $=$\frac{3}{2} of number (1); Number $= 18 \div$\frac{3}{2}$= 12$ (1).	1+1
18	$\angle AOD = 180 - 40 = 140^\circ$ (1); $\angle OAB = (180 - 40) \div 2 = 70^\circ$ (1).	1+1
19	square: 4 (1); regular octagon: 8; total 12 (1).	1+1
20	$-6 - 9 = -15$ (1); $9 - (-6) = 15$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Between neighbours: 6 cm (1); First to third: 12 cm (1); Total length $= 6 \times 3 = 18$ cm (1).	1+1+1
22	H: order 2, lines 2; A: order 1, lines 1; T: order 1, lines 1 (1); Compare each pair (1); Counting complete (1).	1+1+1
23	After first: 100; After second: -150 (1); After deposit: -30 (1); Negative means overdrawn (1).	1+1+1
24	Along the length: 5 squares; along the width: 3 squares (1); Squares $= 15$ (1); Left over area $= 250$ cm^2 (1).	1+1+1
	OR: Filled $=$\frac{2}{3}$\times 6 = 4$ L (1); More needed $= 2$ L (1); Empty fraction $\frac{1}{3}$ (1).	1+1+1
25	$l + b = 20$, so $b = 6$ cm (1); It is not a square (2).	1+2
	OR: Angle between marks $= 360 \div 12 = 30^\circ$ (1); Lines of symmetry $= 12$ for a regular arrangement (1); Turn to mark 5: 150° (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	After first jump: 5 (1); After second: 0 (1); After third: -9 (1); Distance from 0 $= 9$ (2).	1+1+1+2
	OR: Area of big square $= 400$ cm^2 (1); Area of four small squares $= 4 \times 9 = 36$ cm^2 (1); Remaining area $= 364$ cm^2 (1); Perimeter stays the same as the original $= 4 \times 20 = 80$ cm (1); Cutting corner squares does not change the perimeter (1).	1+1+1+1+1
27	Fraction added $=$\frac{3}{4}$-$\frac{1}{4}$= \frac{1}{2}$ (1); This part is 20 litres (2); Capacity $= 20 \div \frac{1}{2} = 40$ litres (2).	1+2+2

	OR: Width $= 8 - 5 = 3$ m (1); Outer diameter $= 16$ m (1); Inner diameter $= 10$ m (1); Check with the compass-drawn figure (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 6–10: **Perimeter and Area, Fractions, Playing with Constructions, Symmetry, The Other Side of Zero** | Class 6 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 9	Height $= 2\times$ area $\div$ base.
2	**(d)** $\frac{3}{7}$	Divide numerator and denominator by their HCF.
3	**(d)** 7	Half of AB.
4	**(c)** 45	$360\div$ spokes.
5	**(d)** 4	Count the positive integers up to b-1.
6	**(c)** 54	Half of the rectangle's area.
7	**(c)** 21	Divide, then multiply.
8	**(a)** 3	A point is inside if its distance from O is less than the radius.
9	**(a)** 20	Lines of symmetry = $360\div$ angle.
10	**(d)** -3	Add step by step.
11	**(b)** 154	Area $= l \times b$.
12	**(d)** $2\frac{11}{12}$	Convert to improper fractions or borrow.
13	**(a)** 34	Opposite sides are equal.
14	**(d)** 225	Multiply.
15	**(d)** -10	Add the opposites.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	New side 12 cm; New area 144 cm^2; old area 100 cm^2 (1); Increase $=$ 44 cm^2 (1).	1+1
17	Common denominator 24 (1); $\frac{3}{4}$, $\frac{2}{3}$, $\frac{5}{8}$ (1).	1+1
18	Rectangle $=$ 70 cm^2; Square $=$ 64 cm^2 (1); Larger: rectangle (1).	1+1
19	Angle between neighbouring vertices $= 360 \div 6 = 60^\circ$ (1); Turn $= 2 \times 60 = 120^\circ$ (1).	1+1
20	Integers: -4, -3, -2, -1, 0, 1 (1); Number $=$ 6 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$6-(-4)=10$ (1); $10+(-9)=1$ (1); $1-5=-4$ (1).	1+1+1
22	Inner lawn: 34 m by 19 m (1); Inner area 646 m^2 (1); Path area $= 1000 - 646 = 354$ m^2 (1).	1+1+1
23	Ploughed $=$\frac{2}{5}+\frac{1}{4}$=$ $\frac{13}{20}$ (1); Remaining $= 1 - \frac{13}{20}$ (1); $= \frac{7}{20}$ (1).	1+1+1
24	Area of square $=$ 64 cm^2 (1); One triangle is one quarter: 16 cm^2 (1); Angles at centre: four right angles, 360° (1).	1+1+1
	OR: Angle $= 360 \div 6 = 60^\circ$ (1); After 4 turns: 240° (1); Turns for full rotation $=$ 6 (1).	1+1+1
25	Start at 0; after 6 down: -6; After 2 up: -4 (1); After 7 down: -11 (1); Total floors moved $=$ 15 (1).	1+1+1
	OR: First area $=$ 108 cm^2 (1); Second area $=$ 140 cm^2 (1); Larger: second, difference $=$ 32 cm^2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	square: 90° (1); regular octagon: 45° (1); Larger angle: square; difference $= 45^\circ$ (1); Total lines $= 4 + 8 = 12$ (2).	1+1+1+2
	OR: Coldest to hottest: -8, -5, -2, 3, 11 (1); Hottest 11, coldest -8 (1); Difference $=$ 19 (1); Sum $= -1$ (2).	1+1+1+2
27	Hall in cm: 1200 by 900 (1); Tiles along each side: 30 and 22 (1); Number of tiles $=$ 660 (1); Cost $=$ ₹16500 (2).	1+1+1+2
	OR: Books: \frac{1}{6}$\times 240 = 40$ (1); Rest $=$ 200 (1); Bag: 80 (1); Left $=$ 120 (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.